SIMATS ENGINEERING

ASSESSMENT TOOL – 1

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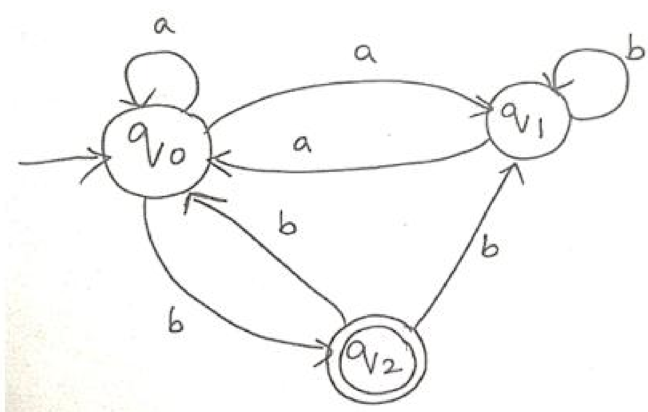
**1.Consider the NFA given below which accepts some language L over the alphabet Σ={a,b}.**

**Construct a DFA that represents the same language L**




ANS :

Given NFA

States: $\{q_0, q_1, q_2\}$

Alphabet: $\Sigma=\{a,b\}$

Initial State: $q_0$

Final State: $q_2$

Transitions

| State | a | b |
|---|---|---|
| $q_0$ | $\{q_0,q_1\}$ | $\{q_2\}$ |
| $q_1$ | $\{q_0\}$ | $\{q_1\}$ |
| $q_2$ | $\{\}$ | $\{q_0,q_1\}$ |

Step 1: Subset Construction

Initial DFA state

$A=\{q_0\}$

Now compute all reachable subsets.

| DFA State | On a | On b |
|---|---|---|
| {qo} | {qo,q1} | {q2} |
| {qo,q1} | {q0,q1} | {q1,q2} |
| {q2} | {} | {q0,q1} |
| {q1,q2} | {qo} | {qo,q1} |
| {} | {} | {} |

Final DFA

Let

$A=\{q_0\}$

$B=\{q_0,q_1\}$

$C=\{q_2\}$

$D=\{q_1,q_2\}$

$E=\emptyset$

Transition Table

| State | a | b |
|---|---|---|
| →A | B | C |
| B | B | D |
| *C | E | B |
| *D | A | B |
| E | E | E |

Final states are those containing $q_2$

Final States = C, D

**2. Let L be the language of words of length atleast 2 whose last two letters are difference from each other. For example abaab is in L but not bb, Construct an NFA for L, Then draw the equivalent DFA**

ANS:

Language:

(ab + ba)

with any prefix.

NFA

States

$q_0$ (start)

$q_1$

$q_2$

$q_3$ (accept)

Transitions

```
q0 --a--> q0
q0 --b--> q0

q0 --a--> q1
q0 --b--> q2

q1 --b--> q3
q2 --a--> q3
```

$q_3$ is final.

Equivalent DFA

Let

A={q0}

B={q0,q1}

C={q0,q2}

D={q0,q1,q3}

E={q0,q2,q3}

Transition Table

| State | A | b |
|---|---|---|
| →A | B | C |
| B | B | D |
| C | E | C |
| *D | B | D |
| *E | E | C |

Final states :

D and E

**3. Design a Nondeterministic Finite Automata (NFA) that takes a binary string as input and accepts the string only if the last two digits are the same (either 00 or 11). Write the formal definition of the NFA. Also show that the string 01011 is accepted by the NFA**

ANS

Formal Definition

Q={q0,q1,q2,q3,q4}

Σ={0,1}

Start=q0

Final={q2,q4}

**Transition Function**

| State | 0 | 1 |
|---|---|---|
| q0 | {q0,q1} | {q0,q3} |
| q1 | {q2} | {} |
| q2 | {} | {} |
| q3 | {} | {q4} |
| q4 | {} | {} |

**State Diagram**

```
            0
      +-----------+
      |           |
  --> q0 ---------+
   | \0         \1
   |  \          \
   |   q1         q3
   |   |0         |1
   |   v          v
   |  q2*        q4*
```

Checking String 01011

Possible path

q0

↓

0

q0

↓

1

q0

↓

0

q0

↓

1

q3

↓

1

q4

$q_4$ is final.

Hence

01011 is Accepted.

**4. Consider the NFA given below which accepts the language representing the set of all strings having 001 as a substring over the alphabet Σ={0,1}. Construct a DFA that represents the same language.**

ANS:

Regular expression

$\Sigma^*001\Sigma^*$

NFA

States

$q_0$

$q_1$

$q_2$

$q_3$(accept)

Transitions

| State | 0 | 1 |
|---|---|---|
| q0 | q0,q1 | q0 |
| q1 | q2 | - |
| q2 | - | q3 |
| q3 | q3 | q3 |

Equivalent DFA

States

A={q0}

B={q0,q1}

C={q0,q1,q2}

D={q0,q3}

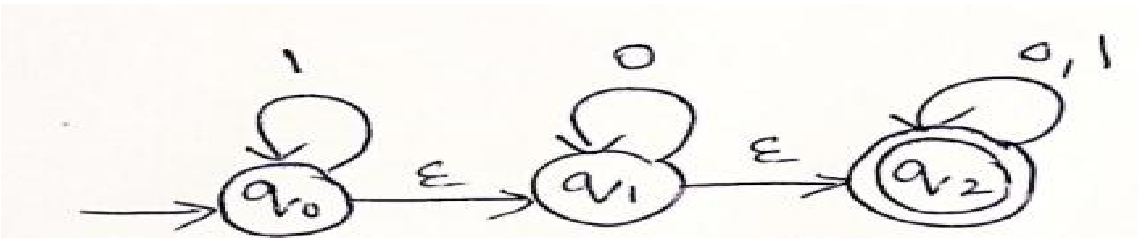
Transition Table

| State | 0 | 1 |
|---|---|---|
| →A | B | A |
| B | C | A |
| C | C | D |
| *D | D | D |

Final State

D

**5. A NFA with ε-transitions is given below. Find ε-closure for each state and Design an equivalent DFA which accepts the same language. Draw its transition table and mark the initial and final states**



ANS:

Given ε-NFA

q0 --ε--> q1 --ε--> q2

Loops

q0 : 1

q1 : 0

q2 : 0,1

$q_2$ is final.

Step 1: ε-closures

ε-closure(q0)

={q0,q1,q2}

ε-closure(q1)

={q1,q2}

ε-closure(q2)

={q2}

Step 2: DFA States

A={q0,q1,q2}

B={q1,q2}

C={q2}

D=∅

**Transition Table**

| DFA State | 0 | 1 |
|---|---|---|
| →*A | B | A |
| *B | C | C |
| *C | C | C |
| D | D | D |

Since every reachable state contains $q_2$, all reachable states are accepting.

Final states

A

B

C